

Collective Robotics – Tutorial 5

Collective Decision-Making: Urn Model, Global Switching, Foraging

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1 The Shared Locust Simulation

Tasks 1 and 2 both use the continuous-space, discrete-time locust model of tutorial 4, task 1. The locusts live on a ring of circumference C , move left ($v = -\text{speed}$) or right ($v = +\text{speed}$), and flip direction in one of two situations: when the majority of the locusts inside their perception range r move the opposite way, or spontaneously with probability P per time step. The switching rules are unchanged from tutorial 4; only the parameters differ. Because both tasks need the same simulation, it lives in a single module, `locust.py`, which both import:

```
1 from locust import simulate
2 L = simulate(N=50, T=120, C=0.5, speed=0.01, r=0.045, p=0.15,
   seed=0)
3 # -> array of left-goer counts L_t, length T + 1
```

Two changes to the tutorial-4 code were necessary.

The ring circumference was hard-coded. The old `step()` computed the wrap-around distance as `np.minimum(distances, 1 - distances)` and seeded the positions with `np.random.uniform(0, 1, n)`. Both silently assume $C = 1$. Tutorial 5 uses $C = 0.5$, which would have placed half of the locusts off the ring and computed the wrong arc distances for the rest.

The simulation was too slow. Task 1 requires 5000 independent runs and task 2 requires runs of 200,000 time steps, which the original per-locust Python loop cannot deliver in reasonable time. The loop is replaced by a pairwise distance matrix and a single matrix–vector product. Since every locust lies inside its own perception range, the self-contribution is removed by subtraction rather than by building an $N \times N$ identity mask:

$$\text{net_dir}_i = \sum_j \mathbb{K}[d_{ij} < r] s_j - s_i, \quad s_j \in \{-1, +1\}.$$

Profiling showed that the $N \times N$ distance matrix dominates the runtime and that in `float64` it spills out of the L2 cache at $N = 150$: the cost per step jumped by a factor of six between $N = 100$ and $N = 150$, far more than the $O(N^2)$ scaling predicts. The matrix is therefore built in `float32`, where the rounding error ($\sim 10^{-7}$) is negligible against $r = 0.045$, while the positions themselves stay `float64` so that the position integration does not drift over millions of steps. A single $N = 50$, $T = 120$ run now takes 6 ms.

Verification. The vectorised majority rule was checked to agree *exactly* with the tutorial-4 implementation on 200 random swarm states, and the distribution of the final left-goer count agrees between the two implementations under a two-sample Kolmogorov–Smirnov test ($p = 0.93$). The spontaneous flip rate was measured at 0.1494 against the nominal $P = 0.15$.

2 Task 1 – Urn Model for the Locust Scenario

Task 1 tests whether the microscopic locust simulation can be described by the macroscopic urn model from the lecture. The full simulation uses $N = 50$ locusts on a continuous one-dimensional ring with $C = 0.5$, speed = 0.01, perception range $r = 0.045$ and spontaneous switching probability $P = 0.15$.

For each of 5000 independent runs, the swarm is first simulated for 100 time steps so that it can settle into an organized collective pattern. During the following 20 time steps we record the one-step change

$$\Delta L = L_t - L_{t-1}.$$

Each observed change is binned by the previous state L_{t-1} . For every possible value $L \in \{0, \dots, 50\}$, the empirical function is estimated as

$$\Delta L(L) = \frac{\text{sum of observed one-step changes from } L}{\text{number of visits to } L}.$$

States that were not visited are ignored during the fitting step. The source code is in `task_1/task_1.py`; the resulting figure is shown in Figure 1.

2.1 Subtask a – The Function $\Delta L(L)$

The roots of the measured function,

$$\Delta L(L^*) = 0,$$

are the fixed points of the averaged one-step dynamics. At such a value L^* , the swarm has no expected tendency to increase or decrease the number of left-going locusts.

The slope at a root determines its stability. If the curve crosses the zero axis with negative slope, $d(\Delta L)/dL < 0$, the fixed point is stable: a small perturbation creates an average change that pushes the swarm back toward L^* . If the curve crosses with positive slope, $d(\Delta L)/dL > 0$, the fixed point is unstable: a small perturbation is amplified and the swarm moves away.

In the locust swarm this has a direct interpretation. The symmetric state $s = 0.5$, or $L = 25$, is the unstable, undecided state. A perfectly divided swarm cannot maintain this balance; any slight majority produces positive feedback and drives the swarm toward one of the two collective directions. The stable attractors lie near the edges of the state space, roughly where about 15% or 85% of the swarm moves left. These are cohesive organized states in which the swarm has committed to a direction. The unstable fixed point at $L \approx N/2$ is the same barrier whose crossing is measured in Task 2.

2.2 Subtask b – Fitting the Swarm Urn Model

The urn model from the lecture, with the scaling constant c introduced by the task sheet, is

$$P_{FB}(s, \varphi) = \varphi \sin(\pi s), \tag{1}$$

$$\Delta s(s) = 4c \left(P_{FB}(s, \varphi) - \frac{1}{2} \right) \left(s - \frac{1}{2} \right), \tag{2}$$

where P_{FB} is the probability of positive feedback, $s = L/N$ is the fraction of left-goers and $\varphi \in [0, 1]$ is the feedback intensity.

The measured data are collected as ΔL against L , while the model is written as Δs against s . Before fitting, the data are therefore converted by

$$s = \frac{L}{N}, \quad \Delta s = \frac{\Delta L}{N}.$$

We then fit Equation (2) with SciPy’s nonlinear least-squares optimizer, using bounds $c \geq 0$ and $0 \leq \varphi \leq 1$. Over the 5000 independent runs, the best-fit parameters are

$$c = 0.1666, \quad \varphi = 1.0000.$$

The fitted value of φ reaches the upper bound, which means the data are best explained by the strongest positive-feedback profile allowed by the model.

2.3 Subtask c – Data, Fit and Probability of Positive Feedback

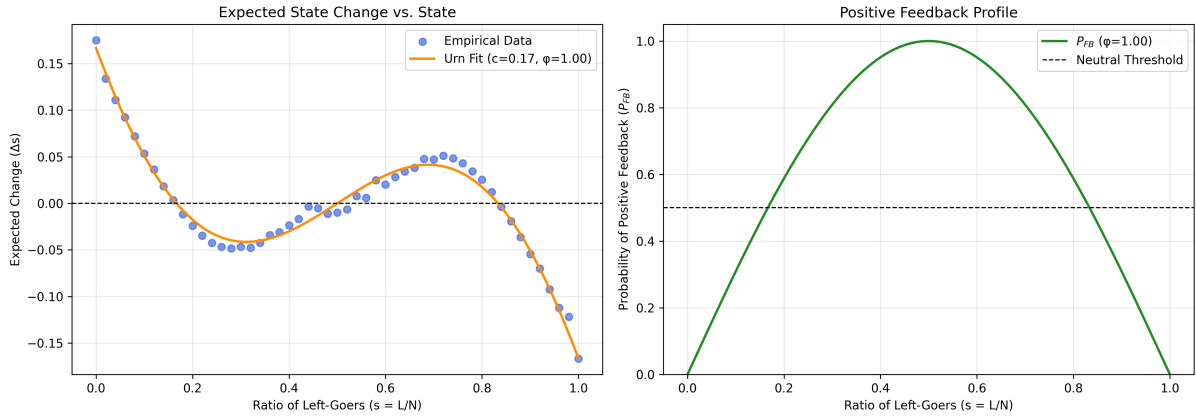


Figure 1: Left: empirical expected state change $\Delta s(s)$ together with the fitted urn model. Right: the resulting probability of positive feedback $P_{FB}(s, \varphi)$ with $\varphi = 1.0$.

Figure 1 shows that the fitted urn model tracks the empirical data closely. The left panel captures the expected sign of the one-step change: near the undecided middle state the dynamics push away from $s = 0.5$, while near the polarized states the dynamics point back toward the stable attractors. This confirms that the simple macroscopic equation captures the main emergent behavior of the microscopic agent-based simulation.

The right panel explains why the middle state is unstable. With $\varphi = 1.0$, the positive feedback probability $P_{FB}(s, \varphi) = \sin(\pi s)$ reaches its maximum value at $s = 0.5$. When the swarm is least organized, individual locusts therefore experience the strongest possible tendency to align with their local neighbors. The spontaneous switching probability of 15 % adds noise, but it does not remove the strong majority-driven feedback that pushes the swarm toward one of the two polarized collective states.

The agreement is not expected to be perfect at every state, because the urn model keeps only the macroscopic variable s and discards the spatial arrangement of locusts on the ring. Still, for the purpose of describing the average one-step dynamics, the fitted urn model is successful: it reproduces the locations and roles of the unstable undecided state and the two stable organized states.

3 Task 2 – Density-Dependent Global Switching

3.1 Overview

In the biological system, a locust swarm that has converged on one direction occasionally switches to the other, and it does so more often at low density. We test whether the simulation of Section 1 reproduces this. We use the task-1 parameter set ($C = 0.5$, speed = 0.01, $r = 0.045$, $P = 0.15$) and vary only the swarm size, $N \in [20, 150]$.

Three zones are defined on the number of left-goers L :

$$\text{Zone A: } L > 0.7N, \quad \text{Zone B: } 0.3N \leq L \leq 0.7N, \quad \text{Zone C: } L < 0.3N.$$

A *global switch* is a traversal of zone B that connects zone A to zone C, or zone C to zone A. While the swarm sits in A or C the counter is zero; entering B starts it; leaving B into the opposite zone stores the counter and resets it; leaving B back into the zone it came from resets the counter without storing it. The source code is in `task_2/task_2.py`.

3.2 Implementation

The zone bookkeeping is implemented as a pure function of the left-goer sequence L_t , which made it possible to test it against hand-constructed sequences whose correct answer is known: a single A→C crossing, a failed crossing that returns to its zone of origin, two crossings in opposite directions, a direct A→C jump with no time spent in B, and an incomplete crossing at the end of a run.

One case is not specified by the task sheet: a run may *begin* inside zone B, in which case there is no zone of origin to compare against. We do not count such a traversal until the swarm has first touched A or C.

Each of the 10 swarm sizes is simulated 10 times for 200,000 time steps, i.e. 2×10^6 time steps per swarm size. This is far more than the 5,000 steps the sheet asks for, and it is necessary: at $N = 150$ a run of 5,000 steps produces no switch at all. All swarm sizes use the same number of time steps, so the switch *counts* are directly comparable across N .

3.3 Two Different Quantities

The counter defined by the task sheet measures the time the swarm spends *inside zone B* during a successful crossing. That is not the same as the time *between* switches, and it turns out that only the latter carries the density dependence. We therefore report both:

- the **crossing time**, i.e. the counter as defined above, and
- the **waiting time**, i.e. the number of time steps between two consecutive switches, measured within each run.

3.4 Results and Discussion

The crossing time is flat. The time spent inside zone B per switch stays at 11–13 time steps for every swarm size (Figure 2, left). This is the quantity the task sheet’s counter measures, and it shows essentially no density dependence. The interpretation is that zone B is not a state the swarm lingers in: once the swarm is disordered enough that

N	Switches	Crossing time	Waiting time
20	16759	11.8 ± 7.4	119 ± 113
30	7100	13.1 ± 6.5	281 ± 247
40	4485	12.6 ± 5.2	445 ± 374
50	3476	11.9 ± 3.9	575 ± 457
65	2094	11.0 ± 2.9	950 ± 815
80	881	11.3 ± 2.4	2252 ± 2261
95	312	10.9 ± 2.1	6211 ± 6839
110	100	11.2 ± 2.0	19001 ± 22705
130	27	11.3 ± 1.9	46127 ± 33865
150	2	12.5 ± 0.5	—

Table 1: Global switching over swarm size. Switch counts are per 2×10^6 time steps (10 runs of 200,000 steps each). Times are mean $\pm$ standard deviation, in time steps. At $N = 150$ only two switches were observed and no single run contained two of them, so no waiting time can be reported.

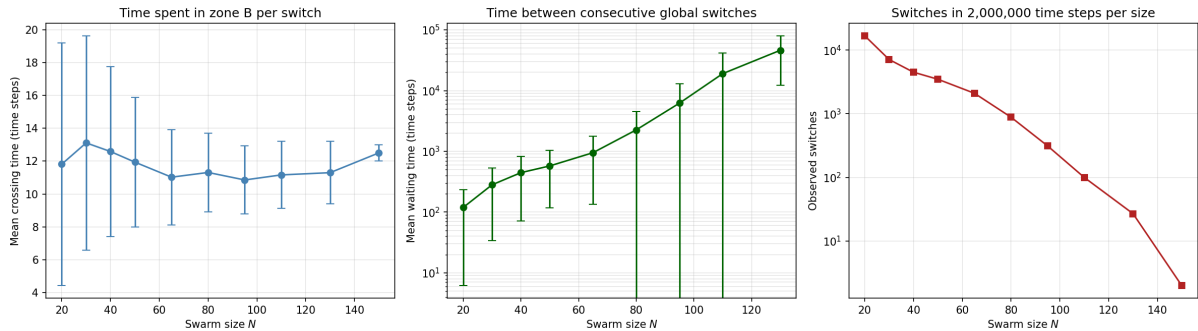


Figure 2: Left: mean crossing time, i.e. the time spent in zone B per switch. Middle: mean waiting time between consecutive switches (logarithmic axis). Right: number of observed switches per swarm size (symmetric-log axis). Error bars are one standard deviation.

$0.3N \leq L \leq 0.7N$, it falls into one of the two consensus attractors in about the same number of steps regardless of how many locusts there are. What changes with density is not how quickly the swarm crosses, but how often it manages to enter the crossing at all.

The waiting time grows steeply. The time between consecutive switches grows from 119 time steps at $N = 20$ to 46,127 at $N = 130$, a factor of about 390 (Figure 2, middle; note the logarithmic axis). Equivalently, the number of switches observed in a fixed budget of 2×10^6 time steps collapses from 16,759 at $N = 20$ to 2 at $N = 150$ (Figure 2, right). The standard deviation of the waiting time is comparable to its mean, as expected for the roughly exponentially distributed waiting time of a rare escape event.

Do we observe the same as in the natural swarm? Yes. A sparse swarm flips its collective direction constantly; a dense swarm locks into whichever direction it happened to choose and almost never leaves it. The mechanism is visible directly in the model. Alignment is driven by the local majority, so a locust surrounded by more neighbours is more strongly pinned to the consensus. A global switch requires enough locusts to defect

at the same time to carry L across the whole of zone B against that pinning, and the only source of defection is the spontaneous flip, which each locust performs independently with probability $P = 0.15$. The probability that a sufficient number of independent flips coincide falls off very quickly as N grows, which is exactly the steep growth of the waiting time observed above.

This is the same escape-from-a-potential-well picture that underlies the urn model of task 1: the unstable fixed point at $L^* \approx N/2$ is the barrier separating the two consensus states, and for large swarms it becomes effectively insurmountable.

4 Task 3 – Foraging

4.1 Overview

We implement a foraging controller for ARGoS3 foot-bots. The robots search a 4×4 m arena for objects, push them into a home zone marked by a bright light, and return to search again. Swarm performance is measured as the number of objects collected in a fixed period, for swarm sizes $N \in \{1, \dots, 10\}$. The simulator configuration, the Lua controller, the C++ loop functions and the analysis scripts live in `task_3/`; the whole toolchain runs inside a single Docker image.

The home zone is a bright yellow light at $(-1.75, 0)$. Forty objects – movable cylinders, each carrying a blue LED so that the robots’ cameras can see them – start in the eastern part of the arena, away from home, so that they have to be transported. Each swarm size was run 20 times for 300 simulated seconds, giving 200 runs in total.

4.2 Implementation

4.2.1 Emulating the bumper

The task sheet specifies a bumper sensor that measures a force from which the robot can tell how many objects it is pushing. ARGoS has no such sensor, so it is emulated from the proximity ring and the omnidirectional camera. We first measured, on this ARGoS build, what an object straight ahead of a robot reads:

Distance to object	16.5 cm	19 cm	25 cm	35 cm
Proximity reading	1.00	0.37	0.11	0.00

Table 2: The proximity ring fires well before the robot is in contact. Contact occurs at 16.5 cm: the foot-bot radius (8.5 cm) plus the cylinder radius (8 cm).

The bumper is therefore *pressed* when the proximity reading exceeds 0.35 (the object is then within ≈ 19 cm) *and* the camera reports an object blob inside the front arc at a distance of at most 18.5 cm. The measured *force* is the number of such blobs, i.e. how many objects the robot is currently pushing. Objects are the only blue things in the arena – no robot state uses a blue LED – so a blue blob is always an object and never a team-mate.

Taking Table 2 seriously matters. An early version treated any blob further than 20 cm as unable to explain a proximity hit, so cylinders at 25 cm – which still register 0.11 on the ring – were reported as walls. The `boundary` flag then fired in the middle of the arena, at a mean distance of 1.18 m from the nearest wall.

4.2.2 Scoring from ground truth

Swarm performance is measured by a small C++ loop function (`foraging_loop.cpp`) rather than from the robots’ own reports. An object counts as collected once its centre enters a disc of radius 0.35 m around the home light, and it is then respawned at a random position in the foraging field.

Respawning is not cosmetic. It keeps the swarm from exhausting the resource during the observation window, and, more importantly, it prevents delivered objects from piling up at the edge of the home zone, where they would sit inside the robots’ camera range and be counted by the bumper as objects currently being pushed. Before respawning was introduced, the measured force reached 7 for a robot pushing a single object, and the reported performance was correspondingly inflated.

4.2.3 The controller

The controller (`foraging.lua`) is a five-state machine:

- **SEARCH** – random walk with obstacle avoidance. Inside the home zone the robot instead walks away from the light and ignores objects, so that it does not re-harvest the nest.
- **APPROACH** – steer towards the nearest object blob until the bumper presses.
- **TRANSPORT** – push towards the light (phototaxis). The obstacle-repulsion vector skips the pushing arc, so walls and team-mates still deflect the robot, but the object it is pushing does not.
- **BACKUP / TURN** – reverse and turn away, after a delivery or after abandoning an object.

Two details turned out to matter.

Delivery is decided on the falling edge of contact. The loop function removes an object as soon as it enters the home disc, which happens while the robot pushing it is still ≈ 0.52 m away. The robot therefore perceives a delivery as “contact lost, next to the light”. The naive rule “no object in the bumper *and* I am near the light” is wrong: a robot that drops an object halfway home keeps coasting towards the light and eventually satisfies it. With that rule the robots over-reported their deliveries by 36 %. Deciding on the tick at which contact is lost – the only moment that carries information about where the object went – reduces the discrepancy to 2 % (Figure 5).

A stuck robot abandons its object. A robot that pushes an object into the wall *beside* the nest would push forever: the object never enters the home disc, so contact is never lost. The `error` output required by the task sheet detects exactly this situation (the robot is commanded to move but does not travel), and we use it: after ten further cycles in that state the robot backs off and resumes searching.

4.2.4 The six required outputs

The controller emits the six values required by the task sheet on every control cycle. The three-valued flags are `true`, `false` or `unknown`; `unknown` occurs for the error flag during the first 30 cycles, while the stuck detector’s history buffer is still filling (1.0 % of all cycles).

A single proximity hit can have more than one explanation at once: a team-mate and an object may both lie near the reported bearing. We therefore believe whichever candidate is *nearer*. Two independent checks confirm that the resulting `collision` flag is meaningful. It never fires at $N = 1$, when there are no other robots to collide with. And when it does

Output	How it is derived
Motor left/right	the commanded wheel velocities
Error	commanded to move, but travelled < 2 cm in 3 s
Collision	the proximity hit is best explained by a RAB neighbour
Arena boundary	the hit is explained by neither a neighbour nor an object
Transporting object	the robot is in state TRANSPORT
Home zone	sum of the 24 light readings > 4.21 (≈ 0.35 m)

Table 3: The six controller outputs and how each is derived from the sensors.

fire, the mean distance to the nearest other robot is 0.19 m, against 0.48 m when it does not – the foot-bot contact distance is ≈ 0.17 m.

4.3 Results

N	Collected	Per robot	Collision	Boundary
1	1.30 ± 0.66	1.30	0.0 %	1.7 %
2	3.30 ± 1.17	1.65	5.1 %	2.6 %
3	4.65 ± 1.95	1.55	11.0 %	3.5 %
4	5.55 ± 1.93	1.39	12.4 %	3.5 %
5	6.90 ± 1.68	1.38	15.3 %	4.5 %
6	8.15 ± 1.90	1.36	18.4 %	4.6 %
7	9.75 ± 1.89	1.39	18.6 %	4.8 %
8	9.40 ± 2.06	1.18	20.3 %	5.7 %
9	11.35 ± 1.76	1.26	20.2 %	5.2 %
10	10.80 ± 2.48	1.08	22.5 %	6.2 %

Table 4: Objects collected in 300 s (mean $\pm$ standard deviation over 20 runs), and the fraction of control cycles in which the collision and boundary flags are raised.

Performance is sublinear in the swarm size. The total number of collected objects grows from 1.3 at $N = 1$ to 10.8 at $N = 10$, but not proportionally: a power-law fit over $N \geq 2$ gives collected $\propto N^{0.78}$ (Figure 3, left). Equivalently, per-robot performance falls from 1.65 objects at $N = 2$ to 1.08 at $N = 10$, a loss of about a third.

The loss is interference. The fraction of control cycles in which a robot is in contact with another robot rises almost linearly with the swarm size, from 0 at $N = 1$ to 22.5 % at $N = 10$ (Figure 4, left). Encounters with the arena boundary also rise, but far more slowly (1.7 % to 6.2 %), which is expected: the arena does not change, only the number of robots in it does. Meanwhile the controller’s time budget is remarkably stable – robots spend ≈ 40 % of their cycles transporting at every swarm size (Figure 4, right). Robots in a larger swarm are therefore not failing to find objects; they are spending their time pushing each other out of the way while they do it. This is the standard interference effect: the arena is a shared, finite resource, and the cost of sharing it grows with the number of robots.

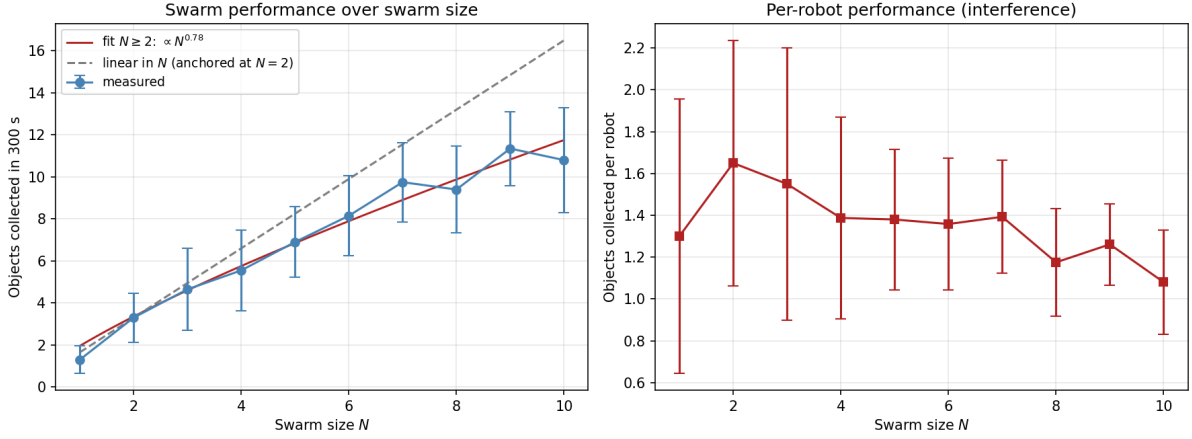


Figure 3: Left: swarm performance over swarm size, with a power-law fit over $N \geq 2$ and a linear reference line anchored at $N = 2$. Right: objects collected per robot.

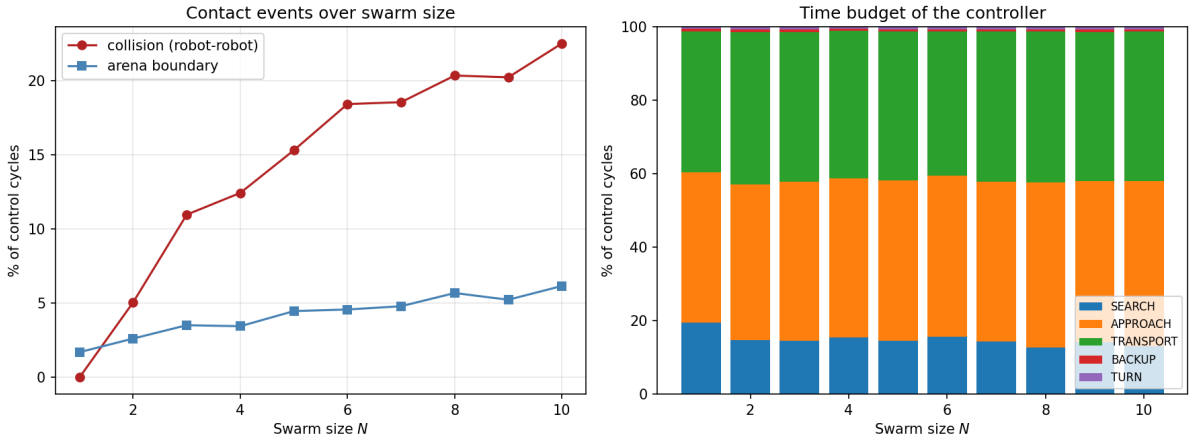


Figure 4: Left: fraction of control cycles reporting a robot–robot collision and an arena boundary. Right: the controller’s time budget.

A single robot is not a swarm of one. Per-robot performance *rises* from $N = 1$ (1.30) to $N = 2$ (1.65) before it begins to fall (Figure 3, right). A lone robot has no teammate to happen upon an object it walked past, and when it gets stuck it loses a large share of its 300 s with nobody to compensate; the variance at $N = 1$ is correspondingly the largest in the sweep. For this reason we did not draw the customary “ideal linear” reference through the single-robot rate, which would place the measured curve *above* the ideal and wrongly suggest superlinear scaling. We fit the exponent instead, and anchor the linear reference at $N = 2$.

Self-reported performance is not trustworthy in general. After the fix described above, the robots’ camera-based delivery count agrees with the ground truth to within 2 % (Figure 5). We nevertheless score performance from the simulator, because that agreement is a property of one carefully chosen light threshold and of the specific geometry of this arena, not something a real swarm could rely on. The general lesson is that a swarm’s own estimate of what it has achieved is a measurement like any other, with its own bias, and it should be validated against ground truth before it is used as a performance metric.

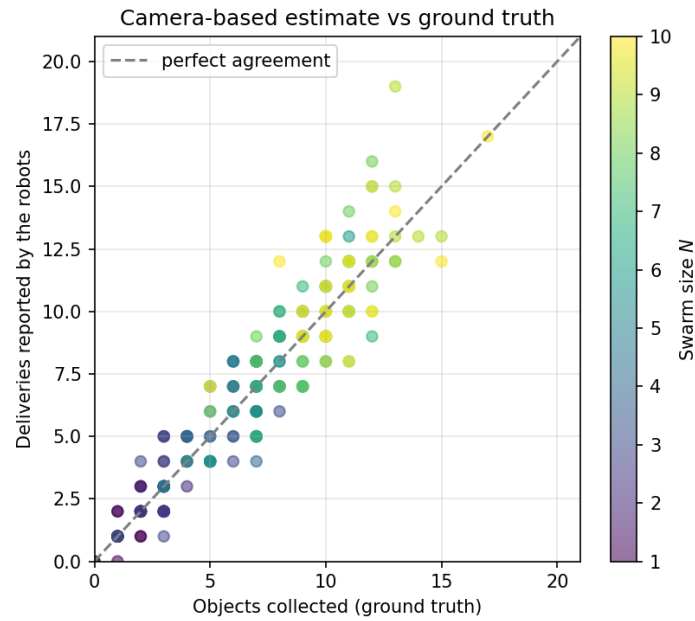


Figure 5: The robots’ own camera-based delivery count against the ground truth measured by the loop functions, one point per run.

4.4 Reproducing the Results

```

1  cd task_3
2  ./run_docker.sh build           # build the ARGoS image (once)
3  ./run_docker.sh loop           # compile the loop functions
4  ./run_docker.sh gui            # watch one simulation
5  ./run_docker.sh experiments    # the full sweep, ~3 min on 8
    cores
6  ./run_docker.sh plot           # write the figures

```

5 Completed Tasks

Task	Description	Status
—	Shared locust simulation, parameterised and vectorised	✓
—	Verification against the tutorial-4 implementation	✓
1	Measurement of $\Delta L(L)$ over ≈ 5000 runs	✓
1	Plot of $\Delta L(L)$; meaning of the zeros L^*	✓
1	Fit of the swarm urn model (c and φ)	✓
1	Plot of data, fit and P_{FB}	✓
2	Zone A/B/C bookkeeping, tested against known sequences	✓
2	Sweep over swarm size (10 sizes, 2×10^6 steps each)	✓
2	Crossing time, waiting time and switch count over N	✓
2	Reproduction of the density-dependent switching effect	✓
3	Foraging controller with the six required outputs	✓
3	Bumper and pushing force emulated from proximity and camera	✓
3	Ground-truth scoring and object respawn (loop functions)	✓
3	Sweep over swarm size $N = 1 \dots 10$ (200 runs)	✓
3	Swarm performance over swarm size; interference analysis	✓